PERSONALIZED LEARNING RECOMMENDATION SYSTEM FOR AN ONLINE LMS

A Product Case Study covering User Stories, Requirements, System Design, Trade-off Analysis, Assumptions and Future Roadmap

INTRODUCTION

Online learning platforms today host large and growing course catalogues, and learners often find it difficult to discover the courses that best match their interests, current skill level, and career goals. This document presents the product design for a Personalized Learning Recommendation System built for an Online Learning Management System (LMS), covering three key user personas: Learners, Instructors, and Platform Administrators.

It walks through the complete product definition process followed for this system: capturing user stories for each persona, defining the functional and non-functional requirements, outlining the core system architecture, evaluating the key product trade-off involved in enabling personalization, and documenting the assumptions made along with a roadmap of future enhancements. The overall objective of this exercise is to design a scalable, secure, and learner-centric platform capable of increasing learner engagement by 25%, improving course completion rates by 30%, and enhancing learner satisfaction by 20%.

STEP - 1

USER STORIES

User stories are short, simple descriptions of a requirement written from the point of view of the end user. They typically follow the format "As a [user], I want to [do something], so that [benefit]," and help the product team stay focused on user value rather than on technical detail. For this LMS, user stories have been captured below for the three primary personas — Learner, Instructor, and Platform Admin — so that the platform is designed around their real, everyday needs.

(a) Learner

1. As a Learner

   I want to login

   So that I can view related options

2. As a Learner

I want to login

So that I can opt for the new course

3. As a Learner

   I want to login

   So that I can submit my project submissions

4. As a Learner

   I want to login

   So that I can view scores

5. As a Learner

   I want to login

   So that I can view feedback

(b) Instructor

1. As an Instructor

   I want to login

   So that I can view dashboard

2. As an Instructor

   I want to login

   So that I can view upload learning material

3. As an Instructor

   I want to login

   So that I can take online classes

4. As an Instructor

   I want to login

   So that I can prepare project topics

5. As an Instructor

   I want to login

So that I can view class schedules

(c) Platform Admin

1. As an Admin

   I want to login

   So that I can create session links

2. As an Admin

   I want to login

   So that I can create profiles for the users

3. As an Admin

   I want to login

   So that I can share the session links

4. As an Admin

   I want to login

   So that I can publish materials to cohort

5. As an Admin

   I want to login

   So that I can send announcements

STEP - 2

FUNCTIONAL REQUIREMENTS

Functional requirements define what the system must do — the specific features, actions, and capabilities that allow each user persona to accomplish their goals on the platform. They are derived directly from the user stories captured above, and describe the core functionalities expected from the Learner, Instructor, and Admin roles, along with the platform-wide features and access modes that support them.

Functionalities:

The functionalities of the **Instructor** include –

- ✓ Signing up for new registration
- ✓ Login and Logout for the already registered user
- ✓ Viewing the course curriculum
- ✓ Viewing the available schedule and confirming out of them
- ✓ Take the Online classes
- ✓ Post the materials, upload the Exam related documents, any recorded videos
- ✓ Prepare the quiz tests / assessments and view the same uploaded by the candidates
- ✓ Evaluate and give the grading individually for the candidate

The functionalities of the **Learner** include –

- ✓ Signing up for new registration
- ✓ Login and Logout for the already registered user
- ✓ Viewing the course curriculum
- ✓ Viewing the available schedule and confirming out of them by subscription
- ✓ Making the payment
- ✓ Attend the Online classes
- ✓ Access the materials, upload the Exam related documents, any recorded videos
- ✓ Attempt the quiz tests / take the assessments
- ✓ View the grading evaluated individually

- ✓ Doubts clarification by mentors
- ✓ Help assistance by the team

The functionalities of the **Admin** include –

- ✓ Creating the profile for new registrations
- ✓ Login and Logout personally
- ✓ Viewing the course curriculum
- ✓ Add/Remove the course
- ✓ Add/Remove the users (Instructor/Candidate)
- ✓ Add/Remove the Materials
- ✓ Display/Generate the Scores and reports
- ✓ Escalate the issues for the necessary team to resolve
- ✓ Report the resolved issues
- ✓ Modifying the schedule and confirming the same to the respective users
- ✓ Create the batches for the online classes
- ✓ Send the notifications if any to the concerned users
- ✓ Post the materials, upload the Exam related documents, any recorded videos
- ✓ Displaying the grading individually for the candidate

Features Ok-ed both by the UX Designer and the Product Owner included-

- User-friendly Interface
- Reports and Analytics
- Responsive design
- Support Services for the users
- Localization and Data Consistency
- Artificial Intelligence

Modes of access – freemium, Subscription and Licensing

Defining Core Functionalities the System Must Support:

User Type	Feature	Description	Priority
Learner	User Registration & Login	Allows learners to create an account, securely log in, and access the learning platform.	High
Learner	Course Discovery & Subscription	Enables learners to browse the course curriculum, choose relevant courses, subscribe, and make payments.	High
Learner	Learning Content Access	Allows learners to attend live classes and access study materials, recorded videos, and other learning resources.	High
Learner	Assessments & Progress Tracking	Enables learners to attempt quizzes, submit assessments, and view grades and performance.	High
Instructor	Course & Content Management	Allows instructors to upload study materials, recorded videos, assignments, and examination documents.	High
Instructor	Assessment & Grading	Enables instructors to create quizzes, evaluate submissions, and assign grades to learners.	High
Admin	User & Course Management	Allows administrators to manage learners, instructors, and courses by adding, updating, or removing them.	High
Admin	Batch & Schedule Management	Enables administrators to create class batches, schedule sessions, and notify users about updates.	Medium
Admin	Reports & Performance Monitoring	Generates learner performance reports, course statistics, and overall platform analytics.	Medium
Admin	Issue & Notification Management	Allows administrators to handle support escalations and send important notifications to learners and instructors.	Medium

STEP – 3

NON – FUNCTIONAL REQUIREMENTS

While functional requirements describe what the system does, non-functional requirements describe how well the system does it — covering qualities such as speed, reliability, security, and scalability. These requirements do not add new features to the product; instead, they set the quality bar the platform must meet so that it remains fast, trustworthy, and able to scale as the learner base grows.

List and explain 1–2 non-functional requirements in each of these categories:

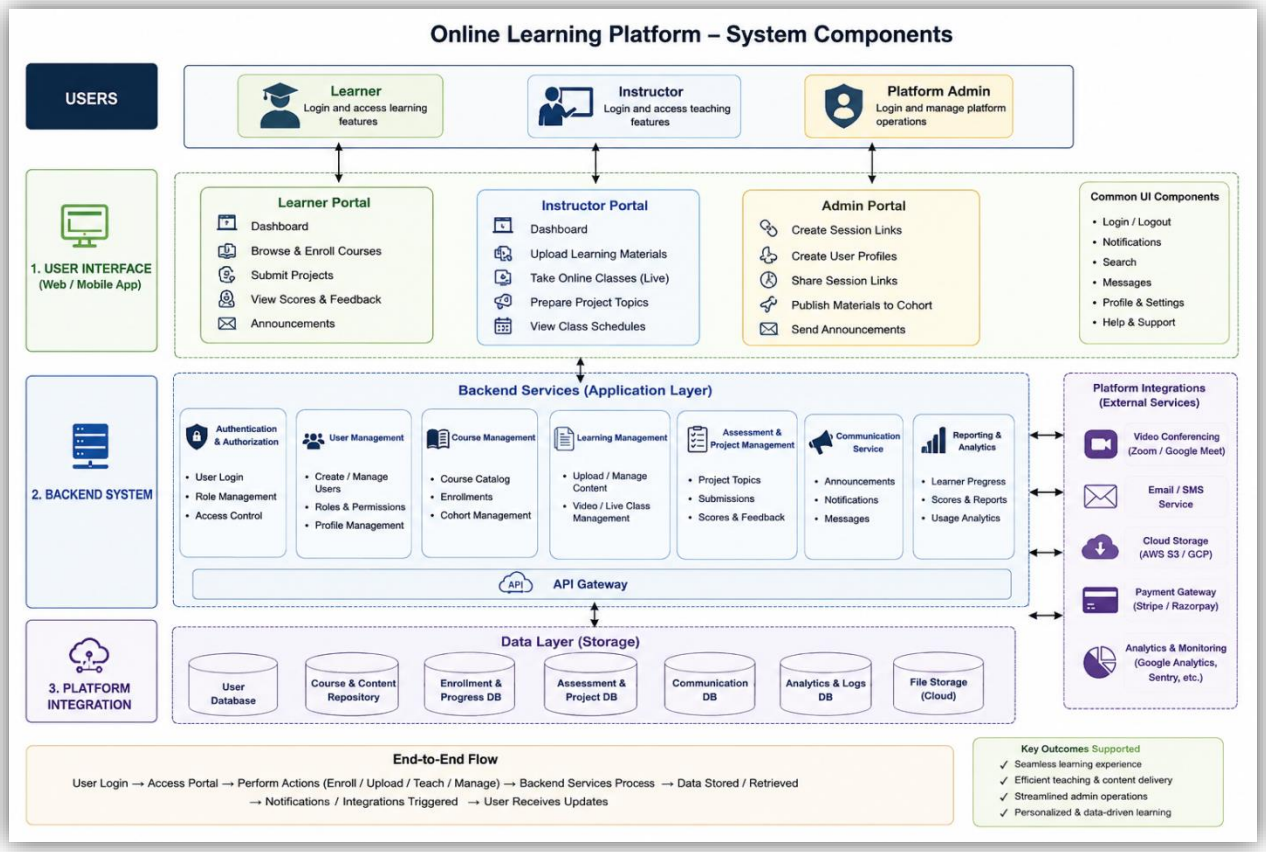
- Performance: The system should provide fast and responsive user interactions, such as loading dashboards, course recommendations, videos, and assessments within an acceptable response time, even during peak usage.
    - Fast Response Time: Course recommendations, dashboards, and search results should load within **2–3 seconds** under normal operating conditions.
    - Efficient Video Delivery: Live classes and recorded videos should stream smoothly with minimal buffering to provide a seamless learning experience.
- Scalability: The system should be capable of supporting a growing number of learners, instructors, courses, and concurrent users without degrading performance.
    - Concurrent User Support: The platform should support thousands of learners and instructors accessing courses simultaneously without performance degradation.
    - Content Scalability: The system should efficiently accommodate an increasing number of courses, videos, assessments, and learning materials as the platform grows.
- Availability: The platform should remain accessible and operational with minimal downtime, ensuring learners can access courses and instructors can conduct sessions whenever required.
    - High System Up Time: The LMS should maintain at least **99.9% availability** so learners can access courses anytime.
    - Fault Recovery: In case of server or network failures, the system should recover quickly with minimal disruption to ongoing learning sessions.
- Compliance: The system should comply with applicable data privacy regulations, copyright policies, and educational standards to ensure legal and ethical handling of user and course data.
    - Data Privacy Compliance: User personal information should be collected, stored, and processed according to applicable data protection regulations.
    - Content Compliance: Uploaded course materials should comply with copyright and intellectual property policies before being published.

- Security & Data Retention: The system should protect user information through secure authentication, authorization, encryption, and secure payment processing while retaining learner records, assessments, and course history according to organizational or regulatory policies.
    - Secure User Authentication: Users should authenticate securely using encrypted credentials and role-based access control for learners, instructors, and administrators.
    - Learning Record Retention: Learner progress, certificates, grades, and assessment history should be securely stored and retained according to the platform's retention policy.
- Monitoring & Logging: The platform should continuously monitor system health, track application performance, log user activities and system events, and generate alerts for errors or failures to enable timely issue resolution.
    - Activity Logging: The system should log important user activities such as logins, course enrollments, assessment submissions, and grading for auditing purposes.
    - System Health Monitoring: The platform should continuously monitor server performance, application errors, and response times, generating alerts when thresholds are exceeded.
- Extensibility: The system should be designed to easily support future enhancements, such as integrating AI-powered recommendation models, new learning formats, third-party tools, payment gateways, or additional educational services without major architectural changes.
    - Modular System Design: The LMS should support the addition of new modules, such as AI-based personalized recommendations or gamification, without major redevelopment.
    - Third-party Integration: The system should be designed to integrate easily with external services such as payment gateways, video conferencing tools, email services, and learning content providers through APIs.

STEP – 4

CORE SYSTEM DESIGN

The core system design translates the functional and non-functional requirements above into a high-level technical architecture. It shows how the Learner, Instructor, and Admin portals interact with the backend application services, how data moves through the system, and how the platform integrates with external services such as payment gateways, video conferencing tools, and cloud storage. The diagram below outlines this end-to-end architecture, from the user interface layer down to the data layer.



STEP – 5

TRADE-OFF ANALYSIS

What is Trade-off Analysis?

Definition (Simple):

Trade-off Analysis is the process of **choosing between two or more options when you cannot have everything at the same time.**

In every product, there are limitations such as:

- Time
- Budget
- Resources
- Technology
- Team size

Key Principle

Trade-off Analysis is **not about finding a perfect solution**. It is about choosing the **best possible solution within real-world constraints**.

A good Product Manager understands that every decision has both:

- **Benefits** (what you gain)
- **Costs or sacrifices** (what you give up)

The goal is to maximize value while accepting the smallest possible downside.

To perform a trade-off analysis, one need to think through these five questions:

1. **What options am I choosing between?**
2. **What do I gain with each option?**
3. **What do I sacrifice with each option?**
4. **Which option best supports the business objectives?**
5. **Why is that trade-off acceptable?**

Major Trade-off in building the system

1. Personalization

What is Personalization?

Personalization means **tailoring the product experience for each individual user** based on who they are and what they do.

Instead of showing everyone the same content, the system recommends content that is most relevant to that specific user.

LMS Example

Suppose three learners log in.

Learner	Interest	Recommendation
Rahul	Data Science	Machine Learning Basics
Priya	UI/UX	Figma for Beginners
John	Cyber Security	Ethical Hacking Fundamentals

Each learner sees different recommendations.

The experience is personalized.

Benefits

- Better engagement
- Higher course completion
- Better learner satisfaction
- Less searching
- More relevant learning

Business Objective Alignment

This directly supports your assignment goals.

✔ Increase Engagement

✔ Improve Completion Rate

✔ Increase Satisfaction

2. Performance

What is Performance?

Performance refers to **how quickly and efficiently the system responds to user actions.**

Users should not wait unnecessarily.

LMS Example

A learner clicks

"Recommended Courses"

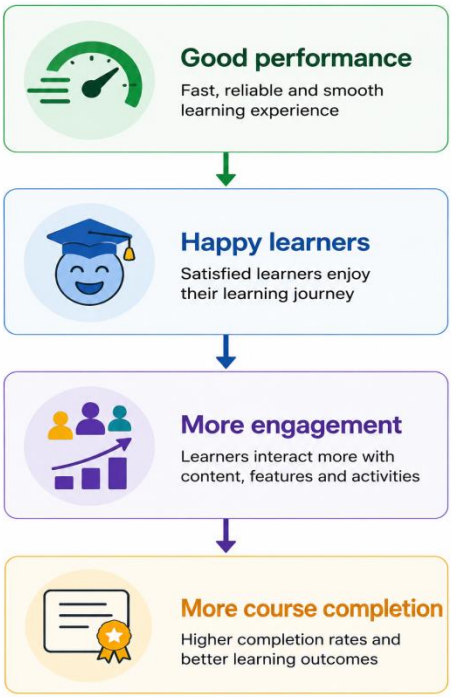
- Good performance
    - Results appear in 2 seconds.
- Poor performance
    - Results appear in 20 seconds.

Performance includes

- Fast login
- Fast search
- Fast recommendations
- Fast video loading
- Quick dashboard loading

Business Impact

Good performance does not just improve speed — it sets off a positive chain reaction across the platform, as shown below:



3. Privacy

What is Privacy?

Privacy means **protecting users' personal information and giving them control over how their data is collected and used.**

LMS Example

The platform stores

- Name
- Email
- Phone number
- Learning history
- Assessment scores
- Career goals
- Skills
- Search history

Privacy ensures

- Only authorized people can access this data.
- Data isn't shared without consent.
- Personal information is protected.

LMS Example

Learner A's scores should not be visible to Learner B.

Only

- Learner
- Instructor
- Admin (if authorized)

should access them.

Benefits

- User trust
- Regulatory compliance
- Secure platform
- Better reputation

4. Tracking

What is Tracking?

Tracking means **collecting user activity data to understand behavior and improve the product.**

Tracking answers questions like

- Which course was viewed?
- Which video was watched?
- Which quiz was attempted?
- Where did users drop off?
- How much time was spent learning?

LMS Example

The example below shows how a single learner’s activity is tracked step-by-step as they move through a course, from logging in to exiting the session:



Now the platform understands learner behavior.

Why is Tracking Important?

Without tracking, the platform doesn't know

- learner interests
- learning habits
- progress
- drop-offs

Without this data, there can be no intelligent recommendations.

Trade-off Analysis for given system

Among the examples given

- Personalization vs Performance
- Privacy vs Tracking

The **most relevant trade-off for current LMS** is: **Privacy vs Tracking**

Why?

It is because entire system depends on AI-powered personalized recommendations.

To recommend the right course, the system needs to know:

- Learning interests
- Completed courses
- Quiz scores
- Skills
- Career goals
- Search history
- Time spent learning
- Course completion patterns

All of this requires tracking user behavior.

What's in Conflict?

Privacy	**Tracking**
Protect user information	Collect user behavior data
Minimize data collection	Collect enough data for AI recommendations
User control	Business insights
Less data stored	More data stored

Both are valuable.

Improving one too much can negatively affect the other.

At the same time, learners expect their personal information to remain private.

Prioritize?

A PM should **not** choose one extreme over the other.

Instead, balance the two by asking:

"What is the minimum amount of data we need to collect to deliver meaningful personalization while respecting user privacy?"

For this LMS, the business objectives are to:

- Increase engagement by 25%
- Improve completion rates by 30%
- Increase satisfaction by 20%

These objectives cannot be achieved without personalized recommendations, which require user behavior data.

Therefore, **tracking should be enabled, but only for data that directly improves learning outcomes**.

Aspect	**Explanation**
Trade-off Identified	Privacy vs. Tracking
What's in Conflict?	Personalized recommendations require tracking learner behavior such as course views, completed lessons, quiz scores, learning preferences, and career goals. However, collecting more user data increases privacy concerns and requires stronger data protection measures.
Priority	Prioritize **tracking that directly supports personalization**, while ensuring user privacy through minimal data collection, transparency, and secure access controls.
Product Decision	The platform will collect only learning-related behavioral data necessary to deliver accurate personalized recommendations and improve engagement. Personally identifiable information and sensitive data will be protected through role-based access, encryption, and clear privacy practices. This balances business growth with user trust.
Reasoning	Personalized recommendations are central to achieving the business objectives of increasing engagement, improving course completion rates, and enhancing user satisfaction. At the same time, maintaining user trust through responsible data handling is essential for long-term platform success.

Why this is the strongest trade-off?

As a Product Manager, one should always connect the trade-off back to the business objectives. In current LMS, the core value proposition is **discovering relevant courses through personalization**. That capability depends on tracking learner behavior, making **Privacy vs. Tracking** the most strategically important trade-off. It demonstrates that we're balancing **business value (better recommendations and engagement)** with **user trust (responsible data usage)**.

STEP – 6

ASSUMPTIONS & FUTURE PLANNING

No product design is complete without acknowledging what has been assumed to be true, and without identifying opportunities for future growth. This section documents the key assumptions made while designing the current system, along with a roadmap of enhancements that can be explored beyond the initial MVP to further strengthen personalization and learner outcomes.

Key Assumptions And Possible Improvements Or Extensions That Could Be Explored In The Future

(A) Assumptions

Key Assumptions

Assumptions are the conditions you believe to be true while designing the product.

Assumption	Explanation
Learners complete their profile during registration.	The recommendation engine assumes users provide information such as interests, skill level, and career goals to generate relevant course recommendations.
Learner activity data is available.	The system assumes learner interactions such as course enrollments, completed lessons, quiz scores, and learning history are captured for personalization.
Courses are properly categorized.	Every course is tagged with metadata such as category, difficulty level, skills covered, and prerequisites to enable accurate recommendations.
Instructors regularly update course content.	The platform assumes instructors maintain updated learning materials, assessments, and recorded sessions to ensure course quality.
Stable internet connectivity is available.	Learners and instructors are expected to have sufficient internet access for attending live sessions and streaming course content.
Secure authentication is implemented.	The system assumes all users are authenticated and authorized using role-based access to protect platform resources and learner data.
Payment services are available.	Subscription-based courses assume integration with a secure payment gateway for successful transactions.
Notifications are delivered successfully.	The platform assumes email, SMS, or in-app notifications reach learners and instructors for schedules, announcements, and reminders.

(B) Future Planning / Possible Improvements

Future planning describes enhancements that can be implemented after the initial MVP.

Future Enhancement	Description
Advanced AI Recommendation Engine	Improve recommendations using collaborative filtering, learning behavior analysis, and predictive AI models that continuously adapt to learner progress.
Learning Path Generation	Automatically generate personalized learning paths that recommend the next best course based on completed topics, skill gaps, and career objectives.
Skill Gap Analysis	Compare learners' current skills with desired career roles and recommend courses to bridge identified gaps.
Gamification	Introduce badges, leaderboards, achievement points, learning streaks, and milestone rewards to increase learner engagement and motivation.
AI Learning Assistant (Chatbot)	Provide an AI-powered assistant that answers learner questions, summarizes course content, recommends resources, and offers 24/7 learning support.
Predictive Dropout Detection	Use analytics to identify learners at risk of dropping out and proactively recommend interventions such as reminders, mentor sessions, or easier learning paths.
Multi-language Support	Support multiple regional and international languages to improve accessibility for a broader learner base.
Integration with Job Portals	Recommend job opportunities and certifications based on completed courses and acquired skills, helping learners transition from learning to employment.
Learning Analytics Dashboard	Provide learners and instructors with detailed insights into progress, completion rates, strengths, weaknesses, and engagement metrics.
Mobile Offline Learning	Enable downloading of course videos and study materials for offline learning, improving accessibility in low-connectivity environments.

SUMMARY

The proposed Personalized Learning Recommendation System for the Online Learning Management System (LMS) has been designed with a user-centric approach by addressing the needs of three primary personas—Learners, Instructors, and Platform Administrators. The solution defines clear user stories and core functionalities that support the complete learning lifecycle, from user onboarding and course enrollment to content delivery, assessments, grading, reporting, and platform administration. In addition, the identified non-functional requirements ensure that the system remains performant, scalable, secure, highly available, compliant, and extensible to support future growth and an enhanced user experience. The high-level system architecture outlines the interaction between the user interface, backend services, and external platform integrations required to deliver these capabilities efficiently. Furthermore, the trade-off analysis demonstrates a balanced product decision by prioritizing personalized learning through responsible user activity tracking while maintaining user privacy and data security, aligning with both business objectives and user trust. Finally, the documented assumptions establish the foundation for the proposed solution, while the future roadmap highlights opportunities such as advanced AI-driven recommendations, personalized learning paths, predictive analytics, gamification, AI learning assistants, and career platform integrations. Overall, the proposed solution effectively supports the business objectives of increasing learner engagement by 25%, improving course completion rates by 30%, and enhancing user satisfaction by 20%, while providing a scalable and future-ready platform capable of evolving with the changing needs of learners and the education ecosystem.